Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<think>
To find the de Bro gl ie wavelength of an electron with a kinetic energy of 1 e V , we need to follow these steps :

1 . Convert the kinetic energy from e V to Jou les : 1 e V = 1 . 602 ÃĹ 10 ^- 19 J , so 1 e V = 1 . 602 ÃĹ 10 ^- 19 J .
2 . Use the kinetic energy formula to find the momentum : kinetic energy = 0 . 5 ÃĹ momentum ^ 2 , rearr ange to get momentum = sqrt (2 ÃĹ kinetic energy)
3 . Convert the momentum from kg *m /s to J *s , this factor having no value in momentum equation works by De Bro gl ie relation .
4 . Use the de Bro gl ie relation : Î» = h / p , where h is the Plan ck constant and p is the momentum .
5 . calculate free length In all formula constants

Plan ck 's constant = 6 . 626 ÃĹ 10 ^- 34 kg *m ^(2)* s ^ (- 1)
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5 . 27 ÃĹ 10 ^- 9 m